

SYDNEY TECHNICAL HIGH SCHOOL



TRIAL HIGHER SCHOOL CERTIFICATE EXAMINATION

2006

MATHEMATICS EXTENSION 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Total Marks - 84

- Attempt Questions 1 - 7
- All questions are of equal value

Name: _____

Teacher: _____

Question 1	Question 2	Question 3	Question 4	Question 5	Question 6	Question 7	Total
/12	/12	/12	/12	/12	/12	/12	/84

Question 1

- a) Find an exact value of $\sin 75^\circ$ 2
- b) Solve $|x-1| > |2-x|$ 2
- c) Find the acute angle between the lines $2x + y = 17$ and $x - y = 3$ (nearest degree) 2
- d) Find the exact value of $\cos (\sin^{-1} \frac{3}{4})$ 2
- e) Solve $\frac{x^2 - 9}{x} \geq 0$ 2
- f) Differentiate $\log (xe^x)$ 2

Question 2

- a) Differentiate $\sin^{-1} (\cos x)$ 2
- b) The roots of $x^2 - 6x + k = 0$ differ by 1. Find the value of k 2
- c) Find $\int \frac{dt}{1+9t^2}$ 2
- d) Evaluate $\int_0^{\sqrt{8}} \frac{x}{x^2+1} dx$, giving your answer in simplest exact form 3
- e) Solve $(\log x)^2 - \log(x^2) = 0$ 3

Question 3

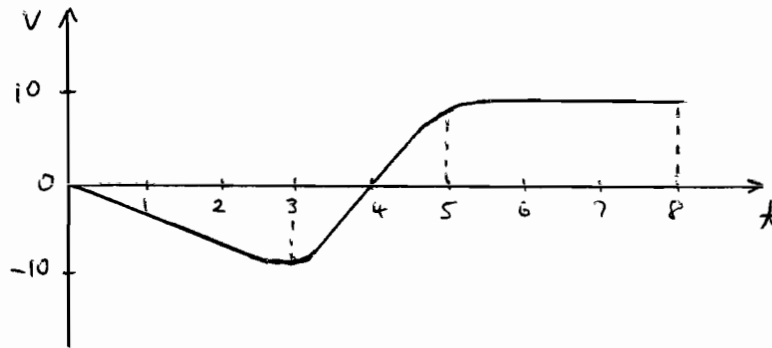
- a) Solve $3^{x-1} = 5$. Give your answer correct to 1 decimal place 2
- b) The equation $x^3 - 3x + 1 = 0$ has a root near $x = 1.5$. Use one application of Newton's Method to find a better approximation for the root, correct to 2 decimal places. 2
- c) The polynomial $P(x) = 4x^3 + kx + 6$ has a factor of $x + 3$. Find the value of k and express $P(x)$ in the form $(x + 3)Q(x)$ 3
- d) (i) Sketch the curve $y = 3 \sin^{-1} 2x$. Clearly indicate values on the axes. 2
- (ii) Find the exact area bounded by the curve $y = 3 \sin^{-1} 2x$, the y axis and 3

Question 4

a) Find $\int \sin^2 4x \, dx$ 2

b) Given $\frac{dx}{dt} = \cos^2 x$ and that $t = 0$ when $x = \frac{\pi}{4}$, find x as a function of t 3

c) The graph below shows the velocity in metres per second of an object for the first 8 seconds.



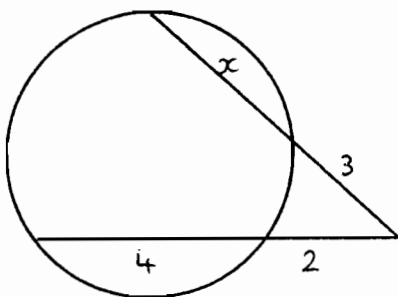
(i) Sketch a graph of its acceleration for $0 \leq t \leq 8$. Do not show units on the vertical axis. 2

(ii) Find a close approximation for the total distance travelled by the object in the first 8 seconds. 1

d) Use the principle of mathematical induction to prove that $3^{2n} - 1$ is divisible by 8 when n is integral and positive. 4

Question 5

a)

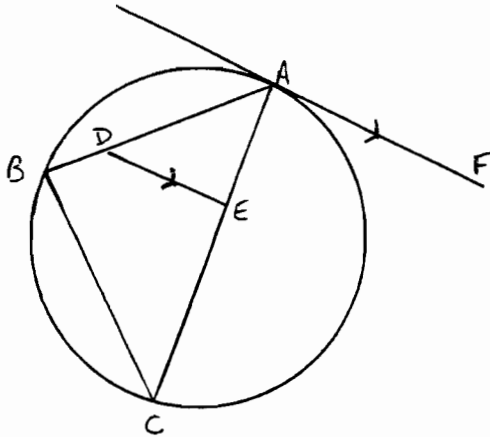


Find x

1

Not to scale

b)

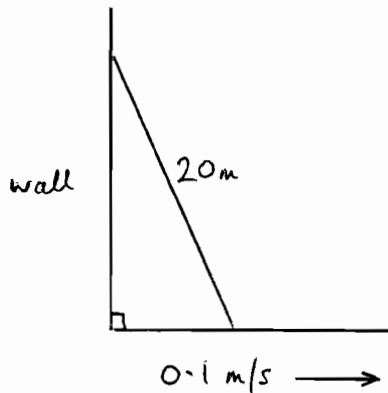


A, B, C are 3 points on the circle
and DE is parallel to tangent AF

3

- (i) Copy this diagram onto your answer page
- (ii) Prove that BDEC is a cyclic quadrilateral

c)



A 20 metre long ladder is resting against a wall. Its base begins to slip along the ground at a rate of 0.1 m/s.

Find the rate at which the top of the ladder is descending when it is 16 metres above the ground.

4

- d) Newton's Law of Cooling states that the rate at which a body loses heat is proportional to the difference between the temperature of the body T and room temperature R .

ie $\frac{dT}{dt} = -k(T-R)$

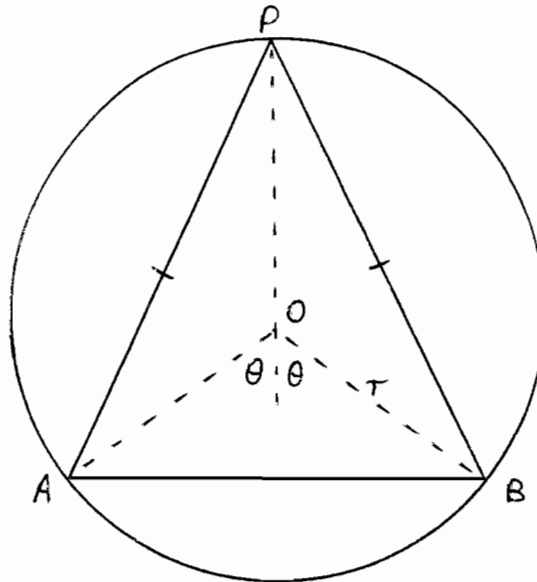
- (i) Show that $T = R + Ce^{-kt}$, where C is a constant, is a solution of this differential equation. 1
- (ii) A cup of coffee cools from 90°C to 50°C in 20 minutes in a room whose temperature is 22°C . Find the temperature of the coffee after 1 hour, to the nearest degree. 3

Question 6

- (a) Solve $\sqrt{3} \cos x - \sin x = 1$ for $0 \leq x \leq 2\pi$ 3
- (b) The speed v m/s of a particle moving along the x - axis is given by
 $v^2 = 24 - 6x - 3x^2$, where x metres is the particle's displacement from the origin.
- (i) Show that the particle is executing Simple Harmonic Motion. 2
- (ii) Find the amplitude and period of the motion. 3
- (c) P and Q are points on the parabola $x^2 = 4ay$ with parameters p and q .
- (i) Find the coordinates of M, the midpoint of PQ. 1
- (ii) If PQ subtends a right angle at the origin, show that $pq = -4$. 1
- (iii) Find the cartesian equation of the locus of M 2

Question 7

(a)



A, P, B are points on a circle, centre O and radius r . Chord AB is such that $AP = BP$ and PO produced bisects $\angle AOB$.

(i) Show that the area of $\triangle APB$ is given by $A = r^2 \sin \theta (1 + \cos \theta)$ 2

(ii) Show that $\frac{dA}{d\theta} = r^2 (\cos \theta + \cos 2\theta)$ 2

(iii) Find the value of θ in radians that will maximize the area of $\triangle APB$ 2

(b) A projectile is fired from a point on a horizontal plane with a velocity of 80 m/s and at an angle of 60° to the horizontal. Take $g = 10 \text{ m/s}^2$.

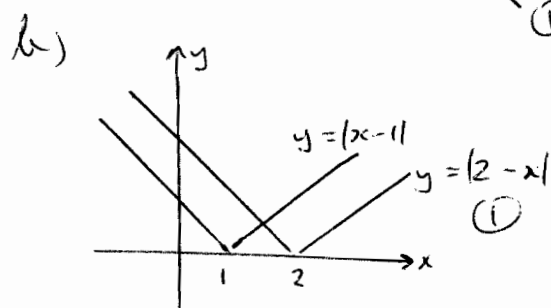
(i) State the vertical and horizontal equations for displacement 2

(ii) Find the time taken for the projectile to reach maximum height 1

(iii) Find the speed of the projectile (to 1 decimal place) and the acute angle to the horizontal (to nearest degree) that the projectile makes one second before impact with the ground. 3

SOLUTIONS (EXT 1 Trial HSC 2006)

1) a) $\sin 75^\circ = \sin(45^\circ + 30^\circ)$ ①
 $= \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$
 $= \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{2}$
 $= \frac{\sqrt{3}+1}{2\sqrt{2}}$ ①



$\therefore x > 1\frac{1}{2}$ ①

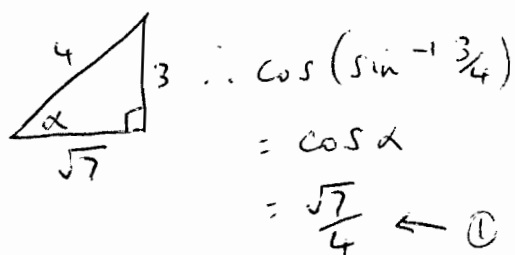
c) $m_1 = -2, m_2 = 1$

$\tan \theta = \left| \frac{-2-1}{1+(-2) \times 1} \right|$ ①
 $= \left| \frac{-3}{-1} \right|$
 $= 3$

$\therefore \theta \doteq 72^\circ$ ①

d) let $\alpha = \sin^{-1} \frac{3}{4}$

$\therefore \sin \alpha = \frac{3}{4}$ ①



2) $\frac{x^2-9}{x} \geq 0 \times x^2 (x \neq 0)$

$\therefore x(x-3)(x+3) \geq 0$ ① method

f) $\log(xe^x) = \log x + \log(e^x)$
 $= \log x + x$ ①

$\therefore \frac{dy}{dx} = \frac{1}{x} + 1$ ①

2)

a) let $y = \sin^{-1} u$ where $u = \cos x$

$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$

$= \frac{1}{\sqrt{1-u^2}} \times (-\sin x)$

$= \frac{-\sin x}{\sqrt{1-\cos^2 x}}$ ①

$= \frac{-\sin x}{\sqrt{\sin^2 x}}$

$= \frac{-\sin x}{|\sin x|} = \pm 1$ ①

b) Roots x and $x+1$

$x+x+1 = -\frac{b}{a}, x(x+1) = \frac{c}{a} = k$

$2x+1 = 6 \therefore 2\frac{1}{2} \times 3\frac{1}{2} = k$

$\therefore x = 2\frac{1}{2}$ ① $\therefore k = 8\frac{3}{4}$ ①

c) $\int \frac{1}{1+9x^2} dx = \int \frac{1}{9(\frac{1}{9}+x^2)} dx$

$= \frac{1}{9} \int \frac{1}{(\frac{1}{3})^2 + x^2} dx$

$= \frac{1}{9} \times \frac{1}{\frac{1}{3}} \tan^{-1}\left(\frac{x}{\frac{1}{3}}\right) + c$

$= \frac{1}{3} \tan^{-1} 3x + c$

$$\begin{aligned}
 d) &= \frac{1}{2} \int_0^{\sqrt{8}} \frac{2x}{x^2+1} dx \\
 &= \frac{1}{2} \left[\log(x^2+1) \right]_0^{\sqrt{8}} \quad \textcircled{1} \\
 &= \frac{1}{2} (\log 9 - \log 1) \\
 &= \frac{1}{2} \log 9 \quad \textcircled{1} \\
 &= \log 3 \quad \textcircled{1}
 \end{aligned}$$

$$\begin{aligned}
 e) &(\log x)^2 - 2 \log x = 0 \\
 \therefore &\log x (\log x - 2) = 0 \\
 \therefore &\log x = 0 \text{ or } \log x = 2 \\
 \therefore &x = 1 \text{ or } e^2 \quad \textcircled{1} \text{ both}
 \end{aligned}$$

$$\begin{aligned}
 3) a) &\log(3^{x-1}) = \log 5 \\
 (x-1) \log 3 &= \log 5 \\
 x \log 3 - \log 3 &= \log 5 \\
 \therefore x &= \frac{\log 5 + \log 3}{\log 3} \\
 &\doteq 2.5 \quad \textcircled{1}
 \end{aligned}$$

$$b) \quad x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\begin{aligned}
 f(1.5) &= 1.5^3 - 4 \cdot 1.5 + 1 \\
 &= -0.125
 \end{aligned}$$

$$\begin{aligned}
 f'(1.5) &= 3(1.5)^2 - 4 \\
 &= 3.75
 \end{aligned}$$

$$\begin{aligned}
 \therefore x_2 &= 1.5 - \left(\frac{-0.125}{3.75} \right) \\
 &\doteq 1.53
 \end{aligned}$$

$$c) \quad P(-3) = 0$$

$$\therefore 4(-27) - 3k + 6 = 0$$

$$\therefore 3k = -102$$

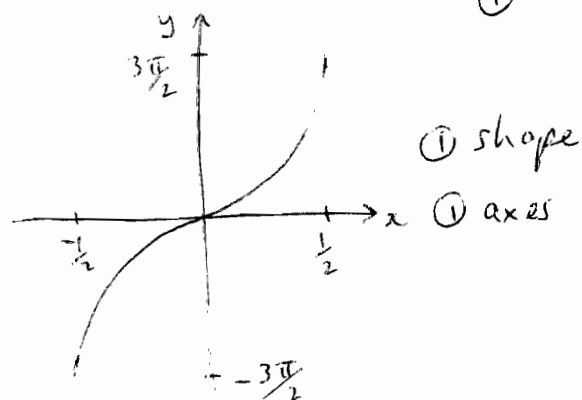
$$\therefore k = -34 \quad \textcircled{1}$$

$$\begin{array}{r}
 4x^2 - 12x + 2 \\
 x+3 \overline{) 4x^3 - 34x + 6} \\
 \underline{4x^3 + 12x^2} \\
 -12x^2 \\
 \underline{-12x^2 - 36x} \\
 2x + 6
 \end{array}$$

① method

$$\therefore P(x) = (x+3)(4x^2 - 12x + 2) \quad \textcircled{1}$$

d) (i)



$$(ii) \quad A = \int f(y) dy$$

$$y = 3 \sin^{-1} 2x$$

$$\frac{y}{3} = \sin^{-1} 2x$$

$$\sin \frac{y}{3} = 2x$$

$$\therefore x = \frac{1}{2} \sin \frac{y}{3} \quad \textcircled{1}$$

$$\begin{aligned}
 \therefore \text{Area} &= \int_0^{3\pi/2} \frac{1}{2} \sin \frac{y}{3} dy \\
 &= \frac{1}{2} \times (-3) \left[\cos \frac{y}{3} \right]_0^{3\pi/2} \quad \textcircled{1} \\
 &= -\frac{3}{2} (\cos \pi/2 - \cos 0) \\
 &= -\frac{3}{2} (0 - 1)
 \end{aligned}$$

Question 4

a) $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$\sin^2 4x = \frac{1}{2}(1 - \cos 8x)$

$\therefore \int \sin^2 4x dx = \frac{1}{2} \int (1 - \cos 8x) dx$
 $= \frac{1}{2} \left(x - \frac{\sin 8x}{8} \right) + c$
 $= \frac{x}{2} - \frac{\sin 8x}{16} + c$

b) $\frac{dt}{dx} = \frac{1}{\cos^2 x}$
 $= \sec^2 x$

$\therefore t = \int \sec^2 x dx$
 $= \tan x + c$

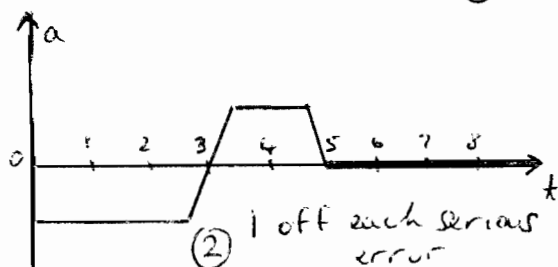
$t=0, x=\frac{\pi}{4} \Rightarrow 0 = 1 + c (c=-1)$

$\therefore t = \tan x - 1$

$\therefore \tan x = t + 1$

$\therefore x = \tan^{-1}(t+1)$

c) i)



(ii) total distance travelled
 $=$ total area enclosed by vel. graph
 $= 15 + 5 + 5 + 30$
 $= 55 \text{ metres}$

d) Prove true for $n=1$:

$3^2 - 1 = 8$ which is divis by 8. $\leftarrow$ ①

Assume true for $n=k$:

i.e. that $3^{2k} - 1 = 8P$ for some integer P . $\leftarrow$ ①

Prove true for $n=k+1$:

i.e. that $3^{2(k+1)} - 1 = 8Q$ for some integer Q .

Now, $3^{2(k+1)} - 1 = 3^{2k+2} - 1$
 $= 3^{2k} \times 3^2 - 1$
 $= 9 \times 3^{2k} - 1$
 $= 9(3^{2k} - 1) + 8$

(from above) $= 9 \times 8P + 8$
 $= 8(9P + 1)$
 $= 8Q$, since $9P + 1$ is integral.

Since the result was true for $n=1$, then from above it must be true for $n=1+1=2$, then $n=2+1=3$ and so on for all positive, integral n .

↑
1 mark off if very poor attempt

Question 5

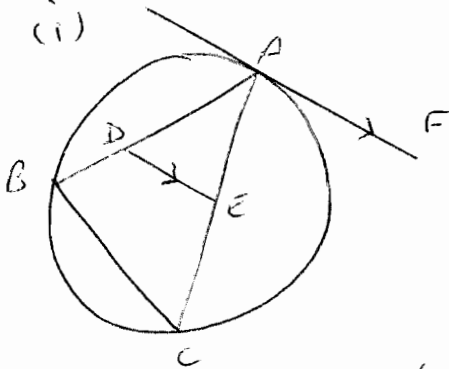
a) $3(3+x) = 2 \times 6$

$$9 + 3x = 12$$

$$3x = 3$$

$$x = 1 \leftarrow \textcircled{1}$$

b) (i)



(ii) $\angle FAE = \angle ABC$ (angle between chord AC and tangent equals angle in alternate segment)

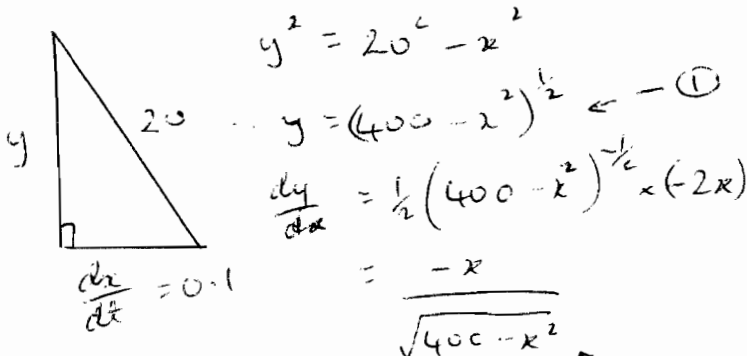
and $\angle FAE = \angle AED$ (alternate angles $AF \parallel DE$)

$$\therefore \angle ABC = \angle AED$$

$\therefore BDEC$ is a cyclic quad.

Since exterior angle equals opposite interior angle.

c)



$$y^2 = 20^2 - x^2$$

$$y = (400 - x^2)^{\frac{1}{2}} \leftarrow \textcircled{1}$$

$$\frac{dy}{dx} = \frac{1}{2} (400 - x^2)^{-\frac{1}{2}} \times (-2x)$$

$$= \frac{-x}{\sqrt{400 - x^2}} \leftarrow \textcircled{1}$$

$$\frac{dy}{dt} = \frac{dy}{dx} \times \frac{dx}{dt}$$

$$= \frac{-x}{\sqrt{400 - x^2}} \times 0.1$$

When $y = 16, x = 12$

$$\therefore \frac{dy}{dt} = \frac{-12 \times 0.1}{\sqrt{400 - 144}}$$

$$= \frac{-1.2}{16}$$

$$= -0.075$$

$\therefore$ ladder is descending at 0.075 m/s .

d) (i) $T = R + Ce^{-kt}$

$$\therefore \frac{dT}{dt} = Ce^{-kt} \times (-k) \quad \textcircled{1} \text{ fully correct method}$$

$$= -kCe^{-kt}$$

$$= -k(R + Ce^{-kt} - R)$$

$$= -k(T - R) \text{ as reqd.}$$

(ii) $t = 0, T = 90, R = 22$

$$\therefore 90 = 22 + Ce^0$$

$$= 22 + C$$

$$\therefore C = 68$$

$$\therefore T = 22 + 68e^{-kt} \quad \textcircled{1}$$

$t = 20, T = 50$

$$\therefore 50 = 22 + 68e^{-20k}$$

$$28 = 68e^{-20k}$$

$$\frac{28}{68} = e^{-20k}$$

$$\log\left(\frac{28}{68}\right) = -20k$$

$$\therefore k = \frac{\log\left(\frac{28}{68}\right)}{-20} \quad \textcircled{1}$$

When $t = 60 \text{ mins}$, $3 \log\left(\frac{28}{68}\right)$

$$T = 22 + 68e$$

$$\approx 27^\circ \quad \textcircled{1}$$

Question 6

$$a) \text{ Let } \sqrt{3} \cos x - \sin x = A \cos(x + \alpha) = 1$$

$$\left. \begin{aligned} A &= \sqrt{3+1} \\ &= 2 \end{aligned} \right\} \text{ ①}$$

$$\therefore \frac{\sqrt{3}}{2} \cos x - \frac{1}{2} \sin x = \cos(x + \alpha) = \frac{1}{2}$$

$$= \cos x \cos \alpha - \sin x \sin \alpha = \frac{1}{2} \quad \text{(ii)}$$

$$\left. \begin{aligned} \cos \alpha &= \frac{\sqrt{3}}{2} \\ \sin \alpha &= \frac{1}{2} \end{aligned} \right\} \therefore \alpha \text{ is in 1st quadrant}$$

$$\therefore \alpha = \frac{\pi}{6} \quad \text{①}$$

$$\therefore \cos(x + \frac{\pi}{6}) = \frac{1}{2}$$

$$\therefore x + \frac{\pi}{6} = \frac{\pi}{3} \text{ (1st, 4th quad)}$$

$$= \frac{\pi}{3} \text{ or } 5\frac{\pi}{3}$$

$$\therefore x = \frac{\pi}{6} \text{ or } 3\frac{\pi}{2} \quad \text{① both}$$

$$b) (i) \ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \quad \text{①}$$

$$= \frac{d}{dx} \left(12 - 3x - \frac{3}{2} x^2 \right)$$

$$= -3 - 3x$$

$$= -3(x+1)$$

$$\left. \begin{aligned} \text{which is in the form} \\ \ddot{x} = -n^2(x-b) \text{ for SHM} \end{aligned} \right\} \text{ ①}$$

$$(ii) n^2 = 3$$

$$\therefore n = \sqrt{3} \quad (n > 0) \quad \text{①}$$

$$\therefore \text{period} = \frac{2\pi}{\sqrt{3}} \text{ secs.}$$

$$\text{amplitude (max } x) \text{ when } v^2 = 0$$

$$\therefore 24 - 6x - 3x^2 = 0 \quad \text{①}$$

$$\therefore x^2 + 2x - 8 = 0$$

$$(x-2)(x+4) = 0$$

$$\therefore x = 2, -4$$

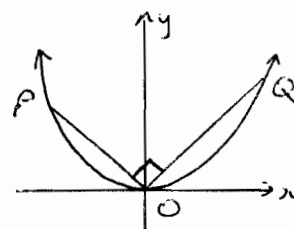
$$\therefore \text{amplitude} = 3 \text{ units.}$$

$$c) (i) P(2ap, ap^2)$$

$$Q(2aq, aq^2)$$

$$M = \left(\frac{2ap+2aq}{2}, \frac{ap^2+aq^2}{2} \right)$$

$$= \left[a(p+q), a\left(\frac{p^2+q^2}{2}\right) \right] \quad \text{①}$$



$$\left. \begin{aligned} M_{OP} &= \frac{ap^2}{2ap}, \text{ similarly} \\ &= \frac{p}{2} \quad m_{OQ} = \frac{q}{2} \end{aligned} \right\} \text{ ①}$$

$$OP \perp OQ \Rightarrow m_{OP} \times m_{OQ} = -1$$

$$\therefore \frac{p}{2} \times \frac{q}{2} = -1$$

$$\therefore pq = -4$$

$$(iii) x = a(p+q), y = a\left(\frac{p^2+q^2}{2}\right)$$

$$\therefore p+q = \frac{x}{a} \Rightarrow y = \frac{a[(p+q)^2 - 2pq]}{2}$$

① method

$$= \frac{a\left[\left(\frac{x}{a}\right)^2 + 8\right]}{2}$$

$$\therefore 2y = \frac{x^2}{a} + 8a$$

$$\therefore y = \frac{x^2}{2a} + 4a \text{ (or } x^2 = 2ay - 8a^2)$$

① answer

Question 7

a) (i) $A = \text{area } \triangle OAP \times 2 + \text{area } \triangle OAB$

$$\textcircled{1} \rightarrow = \frac{1}{2} r^2 \sin(180^\circ - \theta) \times 2 + \frac{1}{2} r^2 \sin 2\theta$$

$$\begin{aligned} \textcircled{1} \rightarrow &= r^2 \sin \theta + \frac{1}{2} r^2 (2 \sin \theta \cos \theta) \\ &= r^2 \sin \theta + r^2 \sin \theta \cos \theta \\ &= r^2 \sin \theta (1 + \cos \theta) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \frac{dA}{d\theta} &= r^2 \cos \theta (1 + \cos \theta) - \sin \theta (r^2 \sin \theta) \\ &= r^2 \cos \theta + r^2 \cos^2 \theta - r^2 \sin^2 \theta \end{aligned}$$

$$\begin{aligned} \textcircled{1} \left\{ \begin{aligned} &= r^2 \cos \theta + r^2 (\cos^2 \theta - \sin^2 \theta) \\ &= r^2 \cos \theta + r^2 \cos 2\theta \\ &= r^2 (\cos \theta + \cos 2\theta) \end{aligned} \right. \end{aligned}$$

(iii) max area when $\frac{dA}{d\theta} = 0$

$$\textcircled{1} \left\{ \begin{aligned} &2 \cos \theta + \cos 2\theta = 0 \\ &\therefore \cos \theta + 2 \cos^2 \theta - 1 = 0 \\ &\therefore 2 \cos^2 \theta + \cos \theta - 1 = 0 \\ &(2 \cos \theta - 1)(\cos \theta + 1) = 0 \end{aligned} \right.$$

$$\therefore \cos \theta = \frac{1}{2} \text{ or } -1$$

$$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \pi$$

$$\therefore \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \pi$$

$\textcircled{1} \rightarrow$ not possible

θ	$\frac{\pi}{6}$	$\frac{\pi}{2}$	$\frac{5\pi}{6}$
$\frac{dA}{d\theta}$	+	0	-

or show that $\frac{d^2A}{d\theta^2} < 0$ when $\theta = \frac{\pi}{6}$

$\therefore$ a maximum area of $\triangle AFB$ occurs when $\theta = \frac{\pi}{6}$

-1 mark if A' or A'' proof method not done

b) (i) $x = vt \cos \alpha$

$$= 80t \times \frac{1}{2}$$

$$= 40t \leftarrow \textcircled{1}$$

$$y = vt \sin \alpha - 5t^2$$

$$= 80t \times \frac{\sqrt{3}}{2} - 5t^2$$

$$= 40\sqrt{3}t - 5t^2 \leftarrow \textcircled{1}$$

(ii) max. height when $y = 0$

$$\therefore y = 40\sqrt{3}t - 5t^2 = 0$$

$$\therefore t = \frac{40\sqrt{3}}{10}$$

$$\textcircled{1} \rightarrow = 4\sqrt{3} \text{ seconds}$$

(iii) time of flight = $8\sqrt{3}$ seconds

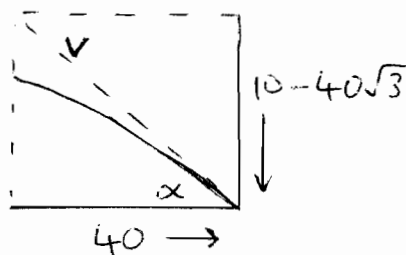
when $t = 8\sqrt{3} - 1$,

$$y = 40\sqrt{3} - 10(8\sqrt{3} - 1)$$

$$= 40\sqrt{3} - 80\sqrt{3} + 10$$

$$\therefore y = 10 - 40\sqrt{3}$$

$$\text{and } x = 40$$



$$v^2 = \sqrt{(10 - 40\sqrt{3})^2 + 40^2}$$

$$\therefore v = 71.5 \text{ m/s} \leftarrow \textcircled{1}$$

$$\text{and } \tan \alpha = \left| \frac{10 - 40\sqrt{3}}{40} \right|$$